

Homework 02

1. Consider a continuous-time sinusoidal wave $x(t) = \cos(2\pi 1000t)$.

(a) This continuous-time signal is converted to the discrete-time signal

$$x[n] = \cos(\omega n) = \operatorname{Re}\{e^{j\omega n}\} \text{ with the sampling frequency } f_s = 8 \text{ kHz.}$$

Find ω , the period N , and $x[n]$ of the discrete-time signal.

n	0	1	2	3	4	5	6	7	8
$x[n]$	1								

(b) This continuous-time signal is converted to the discrete-time signal

$$x[n] = \cos(\omega n) = \operatorname{Re}\{e^{j\omega n}\} \text{ with the sampling frequency } f_s = 4 \text{ kHz.}$$

Find ω , the period N , and $x[n]$ of the discrete-time signal.

n	0	1	2	3	4	5	6	7	8
$x[n]$	1								

(c) This continuous-time signal is converted to the discrete-time signal

$$x[n] = \cos(\omega n) = \operatorname{Re}\{e^{j\omega n}\} \text{ with the sampling frequency } f_s = 2 \text{ kHz.}$$

Find ω , the period N , and $x[n]$ of the discrete-time signal.

n	0	1	2	3	4	5	6	7	8
$x[n]$	1								

(d) This continuous-time signal is converted to the discrete-time signal

$$x[n] = \cos(\omega n) = \operatorname{Re}\{e^{j\omega n}\} \text{ with the sampling frequency } f_s = 1 \text{ kHz.}$$

Find ω , the period N , and $x[n]$ of the discrete-time signal.

n	0	1	2	3	4	5	6	7	8
$x[n]$	1								

2. Consider a continuous-time sinusoidal signal $x(t) = \cos(\omega_a t)$.

$x(t)$ is converted to the discrete-time signal $x[n] = \cos(\omega_d n)$ with the sampling frequency f_s . Find the digital frequency ω_d and fundamental period (N) of the sampled signal in each case,

(1) $\omega_a = 200 \pi$ [rad/sec] , $f_s = 2$ [kHz]

(2) $\omega_a = 500 \pi$ [rad/sec] , $f_s = 1$ [kHz]

(3) $\omega_a = 2,000 \pi$ [rad/sec] , $f_s = 8$ [kHz]

(4) $\omega_a = 2,000,000 \pi$ [rad/sec] , $f_s = 10$ [MHz]

(5) $\omega_a = 4,000,000 \pi$ [rad/sec] , $f_s = 10$ [MHz]

	(1)	(2)	(3)	(4)	(5)
N					
ω_d					